Report Template for IT6007 Object Oriented Design Project

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1. Project Overview

1.1 Project Background

Provide a brief description of the problem or situation that the system is designed to address.

1.2 Problem Statement

Describe the problem that the proposed system aims to solve.

1.3 Project Objectives

List the main objectives of the system.

1. [Objective 1]
2. [Objective 2]
3. [Objective 3]
4. [Objective 4]

1.4 Target Users

Identify the users who will interact with the system.

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2. UML Design

All UML diagrams must be created using **Enterprise Architect (EA)**.

2.1 Use Case Diagram

Insert the completed Use Case Diagram.

2.2 Class Diagram

Insert the completed Class Diagram.

2.3 Sequence Diagrams

Provide sequence diagrams for the **major system operations**.

3. Object-Oriented Design

This section should demonstrate how the project applies the object-oriented concepts learned in the course.

3.1 Encapsulation

Identify an example of encapsulation in the system.

Example Class: [Class Name]

Explanation:

[Explain how data and methods are encapsulated.]

3.2 Inheritance

Identify where inheritance is used.

Parent Class: [Class Name]

Child Class(es): [Class Name(s)]

Explanation:

[Explain why inheritance is appropriate.]

3.3 Polymorphism

Identify an example of polymorphism.

Example:

[Explain how different objects respond differently to the same method/message.]

3.4 Abstraction

Identify an example of abstraction.

Example:

[Explain what implementation details are hidden and what essential features are exposed.]

3.5 Interface

Identify any interface used in the system.

Interface: [Interface Name]

Implementing Class(es): [Class Name(s)]

Purpose:

[Explain why the interface is required.]

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4. User Interface Design

Briefly describe the user interface and explain how it supports the system requirements.

Screen 1 – [Screen Name]

[Insert screenshot]

Screen 2 – [Screen Name]

[Insert screenshot]

Screen 3 – [Screen Name]

[Insert screenshot]

Screen n – [Screen Name]

[Insert screenshot]

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5. Database Design

Insert the database/ERD diagram.

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6. Design Pattern

6.1 Pattern Used

Design Pattern: [Pattern Name]

6.2 Purpose

Explain the problem that the design pattern solves.

6.3 Implementation

Insert the relevant class diagram or code snippet.

[Insert diagram/code]

6.4 Justification

Explain:

- Why the pattern was selected
- Where it is used
- Which classes participate
- How it improves the design

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7. Conclusion

Summarise:

- What was developed
- How the system meets the requirements
- How object-oriented concepts were applied